

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12389588
Kind Code	B2
Date of Patent	August 12, 2025
Inventor(s)	Tang; Yi

Memory and forming method thereof

Abstract

The present disclosure relates to a memory and a forming method thereof. The method of forming a memory includes: forming a stacked layer on a surface of a substrate, the stacked layer including interlayer isolation layers arranged at intervals in a first direction and a sacrificial layer group located between adjacent two of the interlayer isolation layers, the sacrificial layer group including a first sacrificial layer, a second sacrificial layer, and a third sacrificial layer sequentially stacked in the first direction, and the stacked layer including a transistor region, where the first direction is a direction perpendicular to a top surface of the substrate; removing the first sacrificial layer in the transistor region to form a first gap; forming an active pillar in the first gap; removing the second sacrificial layer and the third sacrificial layer in the transistor region to form a second gap.

Inventors:	Tang; Yi (Hefei, CN)
Applicant:	CHANGXIN MEMORY TECHNOLOGIES, INC. (Hefei, CN)
Family ID:	83515100
Assignee:	CHANGXIN MEMORY TECHNOLOGIES, INC. (Hefei, CN)
Appl. No.:	17/815954
Filed:	July 29, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230413512 A1	Dec. 21, 2023

Foreign Application Priority Data

CN	202210703783.4	Jun. 21, 2022
----	----------------	---------------

Publication Classification

Int. Cl.: H10B12/00 (20230101)

U.S. Cl.:

CPC H10B12/30 (20230201); **H10B12/03** (20230201); **H10B12/05** (20230201); **H10B12/482** (20230201);

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
11417659	12/2021	Son	N/A	N/A
2023/0422474	12/2022	Tang	N/A	H10B 12/30

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
112635471	12/2020	CN	N/A

Primary Examiner: Chen; Jack S

Attorney, Agent or Firm: Syncoda LLC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims the priority of Chinese Patent Application No. 202210703783.4, submitted to the Chinese Intellectual Property Office on Jun. 21, 2022, the disclosure of which is incorporated herein in its entirety by reference.

TECHNICAL FIELD

(2) The present disclosure relates to the technical field of semiconductor manufacturing, and in particular, to a memory and a forming method thereof.

BACKGROUND

(3) As a semiconductor device commonly used in an electronic device such as a computer, a dynamic random access memory (DRAM) includes a plurality of memory cells, and each of the memory cells usually includes a transistor and a capacitor. The transistor has a gate being electrically connected to a word line, a source being electrically connected to a bit line, and a drain being electrically connected to the capacitor. A word line voltage on the word line can control on and off of the transistor, such that data information stored in the capacitor can be read through the bit line or data information can be written into the capacitor through the bit line.

(4) Faster response speed, lower power consumption, and higher storage density are always required for a memory such as DRAM. As the size of a memory such as DRAM continues to shrink, the difficulty of a memory manufacturing process and the internal stress of the memory also continue to increase, thereby reducing the yield of the memory. Therefore, how to simplify the memory manufacturing process and improve the memory yield is a technical problem to be solved

urgently at present.

SUMMARY

(5) According to some embodiments, the present disclosure provides a method of forming a memory, including: forming a stacked layer on a surface of a substrate, the stacked layer including interlayer isolation layers arranged at intervals in a first direction and a sacrificial layer group located between adjacent two of the interlayer isolation layers, the sacrificial layer group including a first sacrificial layer, a second sacrificial layer, and a third sacrificial layer sequentially stacked in the first direction, and the stacked layer including a transistor region, where the first direction is a direction perpendicular to a top surface of the substrate; removing the first sacrificial layer in the transistor region, and forming a first gap; forming an active pillar in the first gap; removing the second sacrificial layer and the third sacrificial layer in the transistor region, and forming a second gap; and forming a word line covering a part of the active pillar in the second gap.

(6) According to some other embodiments, the present disclosure further provides a memory, including: a substrate; and a stacked structure located on a top surface of the substrate and including memory cells arranged at intervals in a first direction, where the first direction is a direction perpendicular to the top surface of the substrate; each memory cell includes a transistor and a capacitor electrically connected to the transistor, the transistor includes an active pillar, the capacitor includes a bottom electrode layer, the bottom electrode layer includes a terminal portion and a body portion, and the terminal portion protrudes from the body portion in a second direction and is in contact and electric connection with the active pillar, where the second direction is a direction parallel to the top surface of the substrate.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a flowchart of a method of forming a memory according to a specific implementation of the present disclosure;

(2) FIG. 2 to FIG. 20 are schematic structural diagrams of main processes for forming a memory according to a specific implementation of the present disclosure; and

(3) FIG. 21 to FIG. 23 are schematic structural diagrams of a memory according to a specific implementation of the present disclosure.

DETAILED DESCRIPTION

(4) Specific implementations of a memory and a forming method thereof provided in the present disclosure will be described below in detail with reference to the accompanying drawings.

(5) This specific implementation provides a method of forming a memory. FIG. 1 is a flowchart of a method of forming a memory according to a specific implementation of the present disclosure. FIG. 2 to FIG. 20 are schematic structural diagrams of main processes for forming a memory according to a specific implementation of the present disclosure, where FIG. 2 is a top schematic structural diagram of a memory formed in one embodiment of this implementation, and FIG. 3 to FIG. 20 are any one or two of a top schematic structural diagram of main processes for forming a memory, a schematic cross-sectional diagram of a a-a position in FIG. 2, and a schematic cross-sectional diagram of a b-b position in FIG. 2, to clearly indicate a forming process of the memory. As shown in FIG. 1 to FIG. 20, the method of forming a memory includes:

(6) Step S11, form a stacked layer on a surface of a substrate 30, the stacked layer including interlayer isolation layers 20 arranged at intervals in a first direction D1 and a sacrificial layer group 32 located between adjacent two of the interlayer isolation layers the sacrificial layer group 32 including a first sacrificial layer 321, a second sacrificial layer 322, and a third sacrificial layer 323 sequentially stacked in the first direction D1, and the stacked layer including a transistor region TP, where the first direction D1 is a direction perpendicular to a top surface of the substrate 30, as

shown in FIG. 3.

(7) Specifically, the substrate **30** may be, but is not limited to, a silicon substrate. This specific implementation is described by using an example in which the substrate **30** is a silicon substrate. In other embodiments, the substrate **30** may alternatively be a semiconductor substrate such as a gallium nitride substrate, a gallium arsenide substrate, a gallium carbide substrate, a silicon carbide substrate or an SOI substrate. The substrate is configured to support a device structure thereon. The top surface of the substrate refers to a surface of the substrate **30** towards the stacked layer.

(8) In one example, the interlayer isolation layers **20**, the first sacrificial layer **321**, the second sacrificial layer **322**, and the third sacrificial layer **323** may be alternately stacked on the top surface of the substrate **30** by using a chemical vapor deposition process, a physical vapor deposition process, or an atomic layer deposition process to form the interlayer isolation layers **20** and the sacrificial layer group **32** alternately stacked in the first direction **D1**. The greater the number of the interlayer isolation layers **20** and the sacrificial layer group **32** that are alternately deposited, the greater the storage capacity of the formed memory. There should be a higher etching selection ratio between any two of the first sacrificial layer **321**, the second sacrificial layer **322**, the third sacrificial layer **323**, and the interlayer isolation layers **20**, to facilitate subsequent selective etching. In one example, the etching selection ratio between any two of the first sacrificial layer **321**, the second sacrificial layer **322**, the third sacrificial layer **323**, and the interlayer isolation layers **20** should be greater than 3. In one example, the sacrificial layer group **32** may have a thickness of 30-60 nm in the first direction **D1**, to reserve an enough space for formation of the transistor and the capacitor, and be helpful to control the overall size of the memory finally formed.

(9) To simplify the forming process of the sacrificial layer group **32** and reduce the cost of the sacrificial layer group **32**, in some embodiments, the first sacrificial layer is made of a low-dielectric-constant material, the second sacrificial layer is made of undoped polycrystalline silicon, and the third sacrificial layer is made of silicon dioxide. The low-dielectric-constant material refers to a material with a dielectric constant less than 3. For example, the low-dielectric-constant material may be, but not limited to, one or a combination of two or more of SiOH, SiOCH, FSG (fluorosilicate glass), BSG (borosilicate glass), PSG (phosphosilicate glass), and BPSG (borophosphosilicate glass).

(10) In one embodiment, a substrate isolation layer **31** may also be provided between the substrate **30** and the stacked layer, to isolate the substrate **30** and the stacked layer, and further reduce the stress between the substrate **30** and the stacked layer. The substrate isolation layer **31** may be made of an insulating dielectric material such as oxide (such as silicon dioxide).

(11) Step **S12**, remove the first sacrificial layer **321** in the transistor region **TP**, and form a first gap.

(12) Step **S13**, form an active pillar **120** in the first gap, as shown in FIG. 12.

(13) In some embodiments, the stacked layer further includes a capacitor region **CP** located on a side of the transistor region **TP**. Before the first gap is formed, the method of forming a memory further includes: remove the sacrificial layer group **32** in the capacitor region **CP**, and form a third gap **41**, as shown in FIG. 4; and form a capacitor in the third gap **41**.

(14) In some embodiments, the forming a third gap **41** includes: remove the second sacrificial layer **322** entirely in the capacitor region **CP**, and a part of the first sacrificial layer **321** and a part of the third sacrificial layer **323**, and form a first trench **40** located between a remaining part of the first sacrificial layer **321** and a remaining part of the third sacrificial layer **323**, and the third gap **41** located between adjacent two of the interlayer isolation layers **20** and communicated with the first trench as shown in FIG. 4; and form a first isolation layer **50** filling up the first trench **40**, as shown in FIG. 5.

(15) Specifically, after the stacked layer is formed on the top surface of the substrate **30**, the transistor region **TP**, and the capacitor region **CP** located on the outer side of the transistor region **TP** in the second direction **D2** are defined in the stacked layer. The second direction **D2** is a direction parallel to the top surface of the substrate **30**. Next, the first sacrificial layer **321**, the

second sacrificial layer **322**, and the third sacrificial layer **323** in the capacitor region CP may be etched by using a lateral etching process, and an etching amount of the second sacrificial layer **322** is greater than that of the first sacrificial layer **321** and that of the third sacrificial layer **323**, thereby forming the first trench **40** located between a remaining part of the first sacrificial layer **321** and a remaining part of the third sacrificial layer **323**, and the third gap **41** located between adjacent two of the interlayer isolation layers **20** and communicated with the first trench **40**, as shown in FIG. 4. Thereafter, the insulating dielectric material such as an oxide (such as silicon dioxide) may be deposited in the first trench **40**, to form the first isolation layer **50** filling up the first trench **40**. In one example, the first isolation layer **50** has a width of 30-60 nm in the second direction D2, thereby reducing a parasitic capacitance between the capacitor (especially a bottom electrode layer in the capacitor) and a gate electrode of the transistor while improving the insulating performance of the first isolation layer **50**.

(16) The etching amount of the second sacrificial layer **322** is greater than that of the first sacrificial layer **321** and that of the third sacrificial layer **323** in the etching process. On the one hand, subsequent formation of the first isolation layer **50** for isolating the word line and the capacitor in the first trench **40** is facilitated; and on the other hand, the reduction of electric leakage between the word line and the capacitor is also facilitated.

(17) In some embodiments, the forming a capacitor in the third gap **41** includes: remove a part of the first sacrificial layer **321** below the first isolation layer **50**, and form a second trench **60** communicated with the third gap **41**, as shown in FIG. 6; form a bottom electrode layer filling up the second trench **60** and covering an inner wall of the third gap **41**; form a dielectric layer **221** covering a surface of the bottom electrode layer; and form a top electrode layer **222** covering a surface of the dielectric layer **221**, as shown in FIG. 11.

(18) Specifically, a part of the first sacrificial layer **321** in the capacitor region CP is etched again to form the second trench **60** located between the first isolation layer **50** and the interlayer isolation layer **20**, as shown in FIG. 6. Next, a bottom electrode material (such as tungsten, TiN, or other conductive materials) is deposited in the second trench **60** and on the inner wall of the third gap **41**, to form a terminal portion **70** filling up the second trench **60**, and synchronously form a body portion **71** covering the inner wall of the third gap **41**; and the terminal portion **70** and the body portion **71** jointly constitute the bottom electrode layer of the capacitor, as shown in FIG. 7. The terminal portion **70** and the body portion **71** are in direct contact and electric connection, and the terminal portion **70** protrudes from the body portion **71** in the second direction D2. The terminal portion **70** is lower than the body portion **71** in the first direction D1. The terminal portion **70** is formed and electrically connected to the transistor, which not only contributes to reducing the contact resistance between the capacitor and the transistor, but also ensures the stability of connection between the transistor and the capacitor, thereby further improving the electric performance of the memory.

(19) After the bottom electrode layer is formed, the insulating dielectric material such as an oxide (such as silicon dioxide) is deposited in the third gap **41**, to form a first filling layer **80** filling up the third gap **41**, as shown in FIG. 8. Thereafter, the interlayer isolation layers **20**, the first filling layer **80**, and the body portion **71** of the bottom electrode layer in the capacitor region CP are etched to form a capacitor isolation trench **90** passing through the stacked layer and exposing the substrate **30**, as shown in FIG. 9, and the first filling layer **80** is removed along the capacitor isolation trench **90**. A plurality of capacitor isolation trenches **90** are arranged at intervals in the third direction D3, where the third direction D3 is a direction parallel to the top surface of the substrate **30**, and the third direction D3 intersects with (e.g., is orthogonal to) the second direction D2. Subsequently, the insulating dielectric material such as an oxide (such as silicon dioxide) is filled into the capacitor isolation trenches **90**, to form the capacitor isolation layers **101**, as shown in FIG. 10, for isolating the capacitors arranged at intervals in the third direction. Thereafter, a material with high dielectric constant is deposited on the surface of the body portion **71** of the

bottom electrode layer to form the dielectric layer **221**. A top electrode material (such as tungsten, TiN, or other conductive materials) is deposited on the surface of the dielectric layer **221** to form the top electrode layer **222**, as shown in FIG. **11**.

(20) In some embodiments, the stacked layer further includes a bit line region BP, the capacitor region CP and the bit line region BP are arranged on two opposite sides of the transistor region TP in a second direction D2, and the second direction D2 is a direction parallel to the top surface of the substrate **30**. The forming a first gap includes: remove the first sacrificial layer **321** in the bit line region BP and the transistor region TP, and form the first gap.

(21) Specifically, after the capacitor is formed, the first sacrificial layer **321** in the bit line region BP and the transistor region TP may be removed by using a lateral etching process, to form the first gap located between the interlayer isolation layer **20** and the second sacrificial layer **322**. Next, a channel material may be deposited in the first gap by using a lateral atomic layer deposition process, to form an active layer filling up the first gap. Thereafter, the interlayer isolation layer **20**, the second sacrificial layer **322**, the third sacrificial layer **323**, and the active layer in the bit line region BP and the transistor region TP are etched, to form transistor isolation trenches **130** arranged at intervals in the third direction D3, as shown in FIG. **13**. The bottoms of the transistor isolation trenches **130** expose the substrate **30**, and divide the active layer into the active pillars **120** arranged at intervals in the third direction D3, as shown in FIG. **12**. Subsequently, the insulating dielectric material such as an oxide (such as silicon dioxide) is filled into the transistor isolation trenches **130**, to form transistor isolation layers **21**, as shown in FIG. **14**.

(22) In some embodiments, the forming an active pillar **120** in the first gap includes: deposit a metal oxide material in the first gap, and form the active pillar **120**, as shown in FIG. **12**.

(23) The metal oxide material may be, but not limited to, IGZO (indium gallium zinc oxide). The metal oxide material may be transformed between a conductor and an insulator, and using the metal oxide material to form the active pillar **120** does not require a complex doping process, thereby simplifying the memory forming process.

(24) Step S14, remove the second sacrificial layer **322** and the third sacrificial layer **323** in the transistor region TP, and form a second gap **150**, as shown in FIG. **15**.

(25) Step S15, form a word line **24** covering a part of the active pillar **120** in the second gap **150**, as shown in FIG. **18**.

(26) In some embodiments, the forming a second gap **150** includes: remove the second sacrificial layer **322** and the third sacrificial layer **323** in the bit line region BP and the transistor region TP, and form the second gap **150**, as shown in FIG. **15**.

(27) In some embodiments, the active pillar **120** includes a channel region, and a source region and a drain region arranged on two opposite sides of the channel region in the second direction D2, and the drain region is electrically connected to the capacitor. The forming a word line **24** covering a part of the active pillar **120** in the second gap **150** includes: form a second isolation layer **160** covering an inner wall of the second gap **150**, as shown in FIG. **16**; form an initial word line **161** covering a surface of the second isolation layer **160** in the second gap **150**, as shown in FIG. **16**; and remove the second isolation layer **160** and the initial word line **161** in the bit line region BP, and form a fourth gap **180** between the active pillar **120** and the interlayer isolation layer **20**, the initial word line **161** retained in the transistor region TP being used as a word line **24**, and projection of the word line **24** on the top surface of the substrate covering projection of the channel region on the top surface of the substrate **30**.

(28) Specifically, the second sacrificial layer **322** and the third sacrificial layer **323** in the transistor region TP and the capacitor region CP may be removed by using a lateral etching process, to form the second gap **150**, as shown in FIG. **15**. Next, the second isolation layer **160**, and the initial word line **161** coated with the second isolation layer **160** may be formed in the second gap **150** by using a lateral atomic layer deposition process, as shown in FIG. **16**. The initial word lines **161** extend in the third direction D3, and continuously cover the active pillars **120** arranged at intervals in the

third direction D3. A plurality of initial word lines **161** are arranged at intervals in the first direction D1. The ends of the initial word lines **161** extend out of the transistor region TP in the third direction D3 to be electrically connected to word line plugs **25** (referring to FIG. 2). The ends of the initial word lines **161** extending out of the transistor region TP in the third direction D3 are etched to form a stepped structure, as shown in FIG. 17. The stepped structure refers to any adjacent two of the initial word lines **161** in the first direction D1, and one of the initial word lines **161** closer to the substrate **30** in the third direction D3 protrudes from the other of the initial word lines **161**. After the stepped structure is formed, the second isolation layer **160** and the initial word line **161** in the bit line region BP are etched back to form the fourth gap **180** between the active pillar **120** and the interlayer isolation layer **20**, as shown in FIG. 18. The initial word line **161** retained in the transistor region TP is used as a word line **24**, and projection of the word line **24** on the top surface of the substrate **30** covers projection of the channel region on the top surface of the substrate **30**.

(29) In some embodiments, after the word line covering a part of the active pillar **120** is formed in the second gap **150**, the method of forming a memory further includes: form a third isolation layer **190** filling up the fourth gap **180**, as shown in FIG. 19; remove a part of the third isolation layer **190** and a part of the interlayer isolation layer **20**, and form a fifth gap exposing the active pillar **120** in the bit line region BP; and form a bit line **23** in the fifth gap, as shown in FIG. 20.

(30) Specifically, the insulating dielectric material such as an oxide (such as silicon dioxide) may be deposited in the fourth gap **180** by using a lateral atomic layer deposition process, to form the third isolation layer **190** filling up the fourth gap **180**. Thereafter, a part of the third isolation layer **190** and a part of the interlayer isolation layer **20** may be etched by using a photoetching process to form a fifth gap exposing the active pillar **120** in the bit line region BP; and a bit line material (such as tungsten, TiN, or other conductive materials) is deposited in the fifth gap to form the bit lines **23** arranged at intervals in the third direction D3. The bit lines **23** extend in the first direction D1, and continuously cover the surfaces of the active pillars **120** arranged at intervals in the first direction D1. The active pillars **120** are inserted into the bit lines **23**. On the one hand, the bit lines **23** may be supported by the active pillars **120** to improve the structural stability of the bit lines **23**; and on the other hand, stable electric connection between the active region in the active pillar **120** and the bit line **23** may also be ensured, and the problem of poor contact between the bit line **23** and the transistor is reduced or even avoided.

(31) This specific implementation also provides a memory. FIG. 21 to FIG. 23 are schematic structural diagrams of a memory according to a specific implementation of the present disclosure, where FIG. 21 is a top schematic structural diagram of a memory according to this implementation, FIG. 22 is a schematic cross-sectional diagram of a a-a position in FIG. 21, and FIG. 23 is a schematic cross-sectional diagram of a b-b position in FIG. 21. The memory provided in this specific implementation may be formed by using the method of forming a memory shown in FIG. 1 to FIG. 20. As shown in FIG. 21 to FIG. 23, the memory includes: a substrate **30**; and a stacked structure located on a top surface of the substrate **30** and including memory cells arranged at intervals in a first direction D1, where the first direction D1 is a direction perpendicular to the top surface of the substrate **30**; each memory cell includes a transistor and a capacitor electrically connected to the transistor, the transistor includes an active pillar **120**, the capacitor includes a bottom electrode layer, the bottom electrode layer includes a terminal portion **70** and a body portion **71**, and the terminal portion **70** protrudes from the body portion **71** in a second direction D2 and is in contact and electric connection with the active pillar **120**, where the second direction D2 is a direction parallel to the top surface of the substrate **30**.

(32) Specifically, the terminal portion **70** and the body portion **71** are in direct contact and electric connection, and the terminal portion **70** protrudes from the body portion **71** in the second direction D2. The terminal portion **70** is lower than the body portion **71** in the first direction D1. The terminal portion **70** is formed and electrically connected to the transistor, which not only

contributes to reducing the contact resistance between the capacitor and the transistor, but also ensures the stability of connection between the transistor and the capacitor, thereby further improving the electric performance of the memory.

(33) In some embodiments, the active pillar **120** includes a channel region, and a source region and a drain region arranged on two opposite sides of the channel region in the second direction **D2**, and the terminal portion **70** is in contact and electric connection with the drain region; the stacked structure further includes memory cells arranged at intervals in a third direction **D3**, the third direction **D3** is a direction parallel to the top surface of the substrate **30**, and the second direction **D2** intersects with the third direction **D3**; and the stacked structure further includes word lines **24** arranged at intervals in the first direction **D1**, and the word lines **24** extend in the third direction **D3** and continuously cover the channel regions arranged at intervals in the third direction **D3**.

(34) For example, as shown in FIG. **21** to FIG. **23**, the stacked structure includes a plurality of layers of memory cells arranged at intervals in the first direction **D1**. Each layer of memory cells includes a plurality of memory cells arranged at intervals in the third direction **D3**, and the interlayer isolation layer **20** is arranged between adjacent two layers of memory cells, such that the plurality of memory cells form a three-dimensional stacked layer. Each memory cell includes a transistor and a capacitor. The transistor includes the active pillar **120**, and the active pillar **120** includes a channel region, and a source region and a drain region arranged on two opposite sides of the channel region in the second direction **D2**. The capacitor includes a bottom electrode layer, a dielectric layer **221** covering a surface of the bottom electrode layer, and a top electrode layer **222** covering a surface of the dielectric layer **221**. The bottom electrode layer includes the terminal portion **70** in contact and electric connection with the drain region, and the body portion **71** electrically connected to the terminal portion **70**. The dielectric layer **221** covers a surface of the body portion **71**. “A plurality of” in this specific implementation refers to two or more.

(35) The stacked structure includes a transistor region, and a capacitor region and a bit line region arranged on two opposite sides of the transistor region in the second direction **D2**. The transistor region is configured to form the transistor, and the capacitor region is configured to form the capacitor. The stacked structure further includes the word lines **24** arranged at intervals in the first direction **D1**, and the word lines **24** extend in the third direction **D3** and continuously cover the channel regions arranged at intervals in the third direction **D3**. The word lines **24** are electrically isolated from the capacitor through the first isolation layer **50** and the third sacrificial layer **323** located above the first isolation layer **50**. The ends of the word lines **24** extend out of the transistor region in the third direction **D3** to be electrically connected to the word line plugs **25**. The ends of the plurality of word lines **24** extending out of the transistor region in the third direction **D3** form a stepped structure, to lead out signals of the plurality of word lines **24**. The stepped structure refers to any adjacent two of the word lines **24** in the first direction **D1**, and one of the word lines **24** closer to the substrate **30** in the third direction **D3** protrudes from the other of the word lines **24**.

(36) In some embodiments, the stacked structure further includes: second isolation layers **160** arranged at intervals in the first direction **D1**, located on surfaces of the active pillars **120**, and wrapping the word lines **24**. The second isolation layer **160** located between the word line **24** and the channel region is used as a gate dielectric layer of the transistor.

(37) In some embodiments, the stacked structure further includes: bit lines **23** arranged at intervals in the third direction **D3**, extending in the first direction **D1**, and continuously in contact and electric connection with the source regions arranged at intervals in the first direction **D1**. The bit lines **23** and the word lines **24** are electrically isolated through the third isolation layer **190**.

(38) In some embodiments, at least a part of the source region of the active pillar **120** extends into the bit line **23**.

(39) According to the memory and the method of forming a memory provided by some embodiments of this specific implementation, the stacked layer in which the interlayer isolation layers and the sacrificial layer group are alternately stacked is first formed on the top surface of the

substrate, the first sacrificial layer in the sacrificial layer group is then replaced with the active pillar, and the second sacrificial layer and the third sacrificial layer in the sacrificial layer group are at least replaced with the word line. On the one hand, by forming the sacrificial layer group including the first sacrificial layer, the second sacrificial layer, and the third sacrificial layer, the internal stress of the stacked layer can be reduced, thereby being helpful to increase the stacking height of the stacked layer and the storage capacity of the memory to improve the yield and performance of the memory; and on the other hand, the active pillar and the word line are formed by replacing the sacrificial layers, such that the formation of transistor in the memory does not require a complex epitaxial growth and doping process, and the yield of the memory can be further improved while simplifying the memory manufacturing process. Moreover, the capacitor in the memory provided in some embodiments of the present disclosure includes the bottom electrode layer, the bottom electrode layer includes the terminal portion and the body portion, and the terminal portion protrudes from the body portion in the second direction and is in contact and electric connection with the active pillar, such that the contact resistance between the transistor and the capacitor can be reduced to further improve the electric performance of the memory.

(40) The above described are merely preferred implementations of the present disclosure. It should be noted that several improvements and modifications may further be made by a person of ordinary skill in the art without departing from the principle of the present disclosure, and such improvements and modifications should also be deemed as falling within the protection scope of the present disclosure.

Claims

1. A memory, comprising: a substrate; and a stacked structure located on a top surface of the substrate and comprising memory cells arranged at intervals in a first direction, wherein the first direction is a direction perpendicular to the top surface of the substrate; and each memory cell comprises a transistor and a capacitor electrically connected to the transistor, the transistor comprises an active pillar, the capacitor comprises a bottom electrode layer, the bottom electrode layer comprises a terminal portion and a body portion, and the terminal portion protrudes from the body portion in a second direction and is in contact and electric connection with the active pillar, wherein the second direction is a direction parallel to the top surface of the substrate.
 2. The memory according to claim 1, wherein the active pillar comprises a channel region, and a source region and a drain region arranged on two opposite sides of the channel region in the second direction, and the terminal portion is in contact and electric connection with the drain region; the stacked structure further comprises memory cells arranged at intervals in a third direction, the third direction is a direction parallel to the top surface of the substrate, and the second direction intersects with the third direction; and the stacked structure further comprises word lines arranged at intervals in the first direction, and the word lines extend in the third direction and continuously cover the channel regions arranged at intervals in the third direction.
 3. The memory according to claim 2, wherein the stacked structure further comprises: second isolation layers, arranged at intervals in the first direction, located on surfaces of the active pillars, and wrapping the word lines.
 4. The memory according to claim 2, wherein the stacked structure further comprises: bit lines, arranged at intervals in the third direction, extending in the first direction, and continuously in contact and electric connection with the source regions arranged at intervals in the first direction.
 5. The memory according to claim 4, wherein at least a part of the source region of the active pillar extends into the bit line.
-